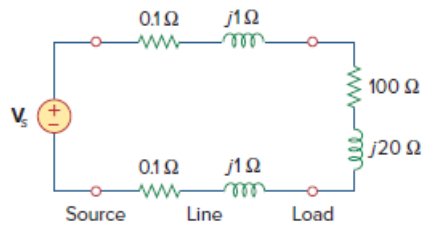


Problem

A power transmission system is modeled as shown in Fig. 11.99. If $V_s = 440\angle 0^\circ$ rms, find the average power absorbed by the load.

Figure 11.99:



Step-by-step solution

Step 1 of 1

$$Z_T = (2)(0.1 + j) + (100 + j20)$$

$$= 100.2 + j22 \Omega$$

$$I = \frac{V_s}{Z_T} = \frac{240}{100.2 + j22}$$

$$P = |I|^2 R_L = 100 |I|^2 = \frac{(100)(240)^2}{(100.2)^2 + (22)^2}$$

$$= 547.3 \text{ W}$$